

G11 PHYSICS

Assignment 02

Due: 21 March 2026

1. Test dimensionally if the formula $t = 2\pi\sqrt{\frac{m}{F/x}}$ may be correct, where t is the time period, m is mass, F is force and x is distance.
2. Young's modulus Y of steel is $19 \times 10^{10} \text{ N/m}^2$. (Young's modulus measures the stiffness of a material and represents the ratio of stress to strain in the elastic region.) Express this value in *dyne/cm*², where *dyne* is the CGS unit of force.
3. Suppose the distance covered by a particle in time t is given by

$$x = a + bt + ct^2 + dt^3.$$

Find the dimensions of a , b , c , and d .

4. An important milestone in the evolution of the universe just after the Big Bang is the set of fundamental scales known as the *Planck units*. These include the Planck time t_p , the Planck length l_p , and the Planck mass m_p . Their values depend on three fundamental constants: (1) the speed of light (the fundamental constant of relativity), $c = 3.00 \times 10^8 \text{ ms}^{-1}$; (2) Newton's gravitational constant (the fundamental constant of gravity), $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$; and (3) Planck's constant (the fundamental constant of quantum mechanics), $h = 6.63 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$.
 - (a) Based on dimensional analysis, find the value of the Planck time t_p . (This represents the smallest meaningful unit of time in physics, often interpreted as the time it takes light to travel one Planck length.)
 - (b) Repeat the derivation to obtain a quantity with the dimensions of length, and evaluate the result numerically. Ignore any dimensionless constants. (This is the Planck length, the smallest meaningful unit of length in physics.)
5. To keep an object moving in a circle at constant speed requires a force called the centripetal force. This force always acts toward the center of the circular path and is responsible for continuously changing the direction of motion. Do a dimensional analysis of the centripetal force.